

The Visual System

Sparse coding

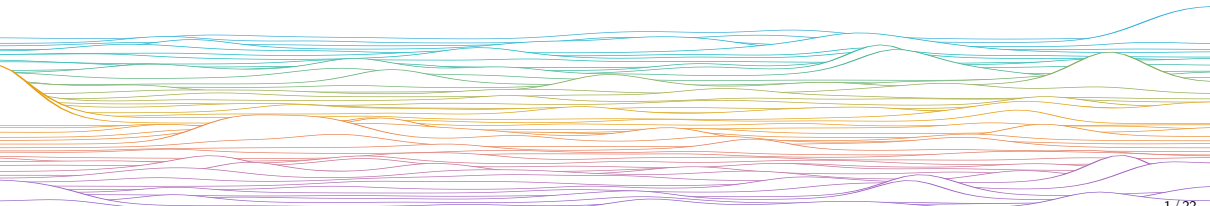
Computational Neuroscience

University of Bristol

mrule

Learning outcomes:

- ▶ Explain role of sparsity in population coding, and interpret associated mathematics.
- ▶ Recognise and discuss biological relevance of Olshausen and Field's 1996 model of V1 simple-cell population codes as emerging automatically when learning a sparse, over-complete basis (or dictionary)



Sparse coding

The activity in some brain regions is **sparse**

- ▶ Many neurons are often silent

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- ▶ 'One-hot' encoding (one neuron active for each stimulus category)

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In V1, sparsity is associated with extracting behaviourally useful visual features efficiently, and in a way that can be used easily by downstream areas.

Sparse coding

Why sparse coding?

- ▶ Balance energy efficiency, representational (coding) capacity?
- ▶ Enhanced discriminability (less overlap between patterns)?
- ▶ Bayesian prior: signals in the world are sparse?

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What do we mean when we say that sparse coding balances energy efficiency and coding capacity?

- ▶ Spikes cost energy
- ▶ Neurons/axons/dendrites cost space/weight/energy to maintain

Sparse Coding in V1
Olshausen and Field (1996)

Sparse coding encourages V1-like receptive fields

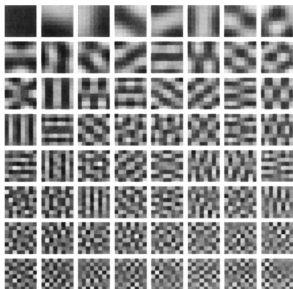
Find set of basis functions that we can linearly sum to reconstruct an image:

$$I_{x,y} = \sum_i x_i \mathbf{W}_{i;x,y}$$

Principal components analysis (PCA) applied to natural images gives tuning unlike V1

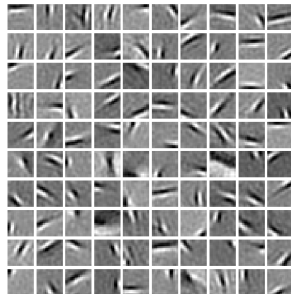
Olshausen & Field (1996): *Iterative Soft Thresholding Algorithm (ISTA)*

PCA



Olshausen and Field, Nature (1996)

ISTA



Olshausen and Field, Nature (1996)

Sparse coding encourages V1-like receptive fields

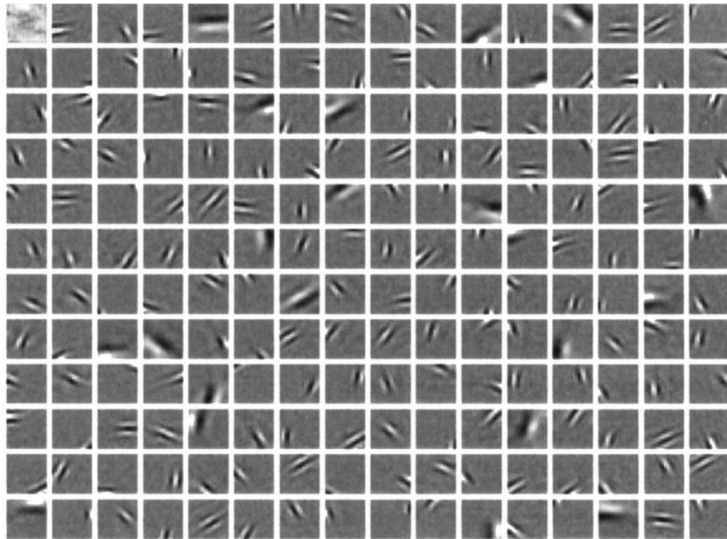
Olshausen and Field searched for basis functions that can be combined to reconstruct images, but also have a sparseness penalty that encourages the x_i terms used to build each image to be small or zero.

Minimise:

$$\text{loss} = \underbrace{\|\mathbf{I}_{x,y} - \mathbf{x}^\top \boldsymbol{\phi}_{x,y}\|_2^2}_{\text{squared error}} + \underbrace{\lambda \sum_i S(x_i)}_{\text{sparsity penalty}}$$

This results in finding a sparse, "over-complete" basis set where components are typically localised, oriented, and bandpass.

Sparse coding encourages V1-like receptive fields



Olshausen and Field, Nature (1996)

Walk-through

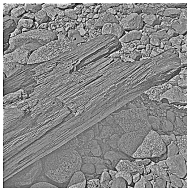
Model retinal center-surround processing

DATASET OF IMAGES



Photoreceptor
pixels

whitening
→

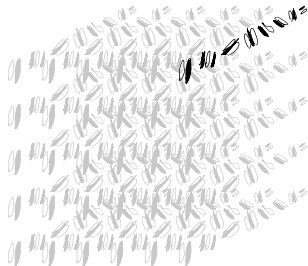


EDGES ENHANCED

RGC

← build a sparse
code to represent
?

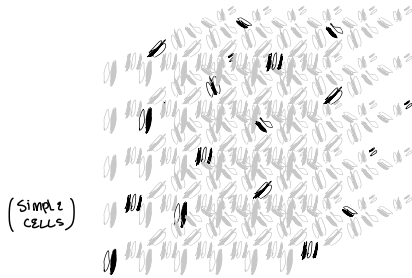
(Simple
cells)



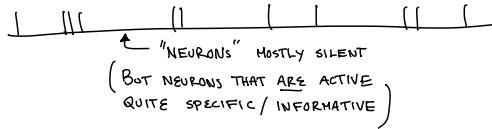
V1 SIMPLE CELLS AS "OVERCOMPLETE" BASIS?
ONE "CELL" FOR EACH POSSIBLE EDGE PATCH?

Represent edges via sparse activation of simple cells

V1 SIMPLE CELLS AS "OVERCOMPLETE" BASIS?
ONE "CELL" FOR EACH POSSIBLE EDGE PATCH?



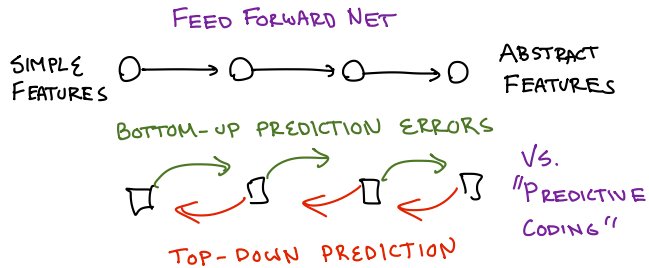
OVERCOMPLETE = MORE DIMENSIONS IN CODE THAN IN PIXELS
BUT, WE WILL CONSTRAIN ACTIVITY TO BE SPARSE



Predictive Coding

Rather than thinking about mapping LGN inputs to simple-cell activity in the forward direction, ask “What sparse activation of simple cells would best match the input?”

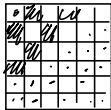
$$\text{loss} = \underbrace{\|\mathbf{I}_{x,y} - \mathbf{x}^\top \boldsymbol{\phi}_{x,y}\|_2^2}_{\text{squared error}} + \underbrace{\lambda \sum_i S(x_i)}_{\text{sparsity penalty}}$$



In biology, everything important is done in closed-loop

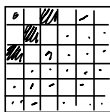
IMAGE INPUT
ARRIVING FROM
RETINA / LGN

TRUE
 y^*



WHITENED
NATURAL IMAGE
- EDGES ENHANCED
- $\approx$ RGC

PREDICT
 $\hat{y}$



VECTOR $\hat{y}$
RE-ORDERED AS
PIXELS "Vec(.)"

PREDICTION



$\hat{y}$

$$\hat{y} = \underline{W} \underline{x}$$

$$\hat{y}_k = \sum_l W_{kl} x_l$$

RECONSTRUCTION
WEIGHTS

V1



$\underline{x}$

SPARSE!
Most neurons silent
 $\sim$ most $x_j \approx 0$

RECONSTRUCTION
ERROR

REG STRENGTH

SPARSE PENALTY

$$\text{Loss}(\underline{W}, \underline{x}, y^*) = \frac{1}{2} \|y^* - \underline{W} \underline{x}\|^2 + \lambda S(\underline{x})$$

pause

Olshausen and Field sparse coding summary

Principal components analysis decomposition of images leads to a _____ in which most neurons are _____ when encoding a specific image. This (is/is not) _____ biologically realistic.

Olshausen and Field found that V1-simple-cell-like population codes emerge naturally when learning a _____ representation.

Olshausen and Field's loss function balanced squared-error _____ with a sparse _____ that encouraged most neurons to take activity close to zero.

_____ is the perspective that sensory neuronal activity represents beliefs about the latent causes of sensory inputs, and uses prediction or reconstruction errors to update these beliefs.

Olshausen and Field sparse coding summary

Principal components analysis decomposition of images leads to a dense code in which most neurons are active when encoding a specific image. This (is/is not) is not biologically realistic.

Olshausen and Field found that V1-simple-cell-like population codes emerge naturally when learning a sparse coding, over-complete basis (or dictionary) representation.

Olshausen and Field's loss function balanced squared-error reconstruction loss with a sparse prior or penalty that encouraged most neurons to take activity close to zero.

Predictive coding is the perspective that sensory neuronal activity represents beliefs about the latent causes of sensory inputs, and uses prediction or reconstruction errors to update these beliefs.

https://githubtocolab.com/engmaths/SEMT30003_2025/blob/main/Labs/lab7/vision_sparse_coding.ipynb

end

